

* Galaxy Morphologies: (i) - (iv)

• looking at Luminosity of these galaxies

• Surface brightness: "Flux"

$$\langle I(r) \rangle = I(r) = I_0 e^{-R/R_h} \quad (\text{Spirals})$$

$$L = \int_0^{2\pi} d\phi \int_0^R I(R) R dR$$

$$u = \frac{1}{R_h}$$

$$\hookrightarrow \int_0^{2\pi} d\phi = \phi \Big|_0^{2\pi} = \underline{2\pi}$$

• Spirals:

$$L = \int_0^{2\pi} d\phi \int_0^\infty I(R) R dR$$

$$= \underline{2\pi I_0 R_h^2}$$

* If, instead wanted

$$L = \int_0^{2\pi} d\phi \int_0^R I(R) R dR$$

• you'll be given the integral

$$* \int_0^{R'} x e^{-ax} dx \quad *$$

$$L = 2\pi \int_0^{R'} I_0 e^{-R/R_h} R dR$$

$$= 2\pi I_0 \int_0^R R e^{-R/R_h} dR$$

$$u = \frac{1}{R_h}$$

• Ellipticals:

-> Different profile
de Vaucouleurs:

$$I(R) = I_e \exp \left[-7.67 \left(\left(\frac{R}{R_e} \right)^{7/4} - 1 \right) \right]$$

$7/4$: profile

$$\leadsto L_{ell} = \int_0^{2\pi} d\phi \int_0^\infty I_0 e^{-7.67 \left(\left(\frac{R}{R_e} \right)^{7/4} - 1 \right)}$$

$$= 7.22 \pi I_e R_e^2$$

$$L_{ell} = (7.22) \pi I_e R_e^2$$

- More general: $1/n$ law:

$$I(R) = I_e e^{-2.67 \left(\left(\frac{R}{R_e} \right)^{2/n} - 1 \right)}$$

$n=1$: exp law (spiral)

$n=4$: elliptical de V.

* CAS classification: send extra notes

- Galactic content:

→ Terminology: Ellipticals "early type"
Spirals "late type"

* Observing: Magnitudes

$$\begin{aligned} \rightarrow \frac{I_1}{I_2} &= 100^{\frac{1}{5}(m_2 - m_1)} \\ \hline m &= \mu \end{aligned}$$

$$\mu(R) = -2.5 \log(I(R))$$

PS: $\mu(R)$: spiral $I(R) = I_0 e^{-R/R_h}$
plug in

$$\begin{aligned} \mu(R) &= -2.5 \log(I_0 e^{-R/R_h}) \\ &= \underbrace{-2.5 (\log_{10} I_0)}_{\mu_0} - 2.5 \log_{10}(e^{-R/R_h}) \\ &= \mu_0 + \log_{10}(e^{-R/R_h}) \end{aligned}$$

• change of base: $\log_a(b^x) = \frac{\log_b(b^x)}{\log_b(a)}$

$$\log_{10}(e^{-R/R_h}) = \frac{\ln(e^{-R/R_h})}{\ln(10)} = \frac{-R/R_h}{\ln(10)}$$

$$\mu(R) = \mu_0 + \underbrace{\frac{2.5}{\ln(10)}}_{1.086} \frac{R}{R_h}$$

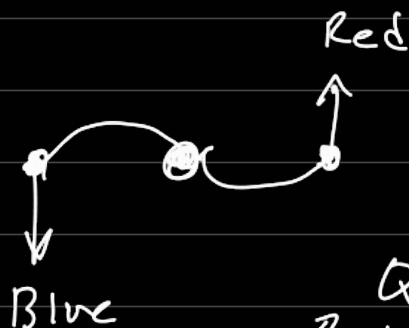
$$\mu(R) = \mu_0 + 1.086 \frac{R}{R_h}$$

* Dynamics: These different velocity profiles

→ Spiral head on

$$\frac{\Delta \lambda}{\lambda} = \frac{V}{c}$$

$$\underline{V(R)} \sim \frac{\Delta \lambda c}{\lambda}$$



QM
 $|1/2\rangle \rightarrow |-1/2\rangle$

- HI 2cm: "outer" spin flip
 Hd: "inner" $n=3 \rightarrow n=2$

- $\left(\frac{M}{L}\right)$: want to find these.

- $\rho(R)$ as well as $n(R)$ and $\Sigma(R)$

- $\left[\frac{M}{L}\right] \sim 4-20$ (spirals)

→ Vir. thm: $2K + U = 0$

- Ellipticals: much more unorganised

→ want to look at

$$\sigma_r^2 = \underline{V_r^2}$$

$$\rightarrow K = \frac{1}{2} \sum_i \underline{m_i} \underline{V_i^2} = \frac{1}{2} M_{gal} \langle \underline{V}^2 \rangle$$

$$\langle \underline{V^2} \rangle = \langle \underline{V_x^2} + \underline{V_y^2} + \underline{V_z^2} \rangle = 3 \langle \underline{V_r^2} \rangle = 3 \sigma_r^2$$

$$K = \frac{3}{2} M_{gal} \sigma_r^2$$

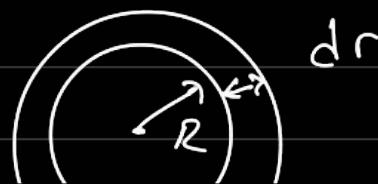
| σ_r : measured line of sight dispersion

- Potential: Newton: $U(R) = - \frac{G M(R) m_{\leftarrow}}{R}$

$$dU = - \frac{G M(R) dm_{\leftarrow}}{R}$$

$$dm = 4\pi r^2 \rho(r) dr \quad (\text{sphere})$$

$$\int dU = - \int_0^R \frac{M(r) G 4\pi r^2 \rho(r) dr}{r}$$



$$U = -\frac{3}{5} \frac{G M_{gal}}{R} \quad (\text{sphere})$$

$$U = -\alpha \frac{G M_{gal}}{R} \quad (\text{in general})$$

• Sphere: $2K + U$: plug in $K = \frac{3}{2} M_{gal} \sigma_r^2$
 $U = -\frac{3}{5} \frac{G M_{gal}}{R}$

$$M_{gal} \approx \frac{5 R \sigma_r^2}{G} \quad (\text{sphere})$$

$$\sigma_r \approx 200 - 300$$

$$\left[\frac{M}{L} \right] \leftarrow \text{total}$$

$$\left[\frac{M}{L} \right] \leftarrow \text{total luminosity}$$

$$\frac{M_{gal}}{L_{ell}} \approx 7 - 14$$

* Unfortunate: $\mu = \left[\frac{M}{L} \right] \approx \frac{M}{L_{un.}}$

earlier we saw $\mu(R) \leftarrow M_{gal}/\text{arcsec}^2$

• so: be careful! and look at units

* Want to look at both spiral & ell:

i) Spiral:

Max amount of luminous output

$$* L = 2\pi I_0 R^2$$

⊗ Idea: assume output is close to this, so that
in general

$$L(R) \approx I_0 R^2$$

C_i : constants

$$\rightarrow L(R) = C_1 I_0 R^2$$

$$R = \frac{L^{1/2}}{C_1^{1/2} I_0^{1/2}}$$

ⓑ Assumption 2: Virial

$$M(R) = C_2 \frac{R v^2}{G}$$

divide by L

$$\mu = \frac{M}{L} = C_2 R v^2$$

$$L = \frac{c_2 R V^2}{\mu G}$$

L as a function of V_{el}

$$L = \frac{c_2 \left(\frac{L^{1/2}}{c_1^{1/2} I_0^{1/2}} \right) V^2}{\mu G} = \frac{c_3 L^{1/2} V^2}{I_0^{1/2} \mu G}$$

$$L = \frac{c_3 V^2}{I_0^{1/2} \mu G} \Rightarrow L(V) = \frac{c_3^2 V^4}{I_0 \mu^2 G^2}$$

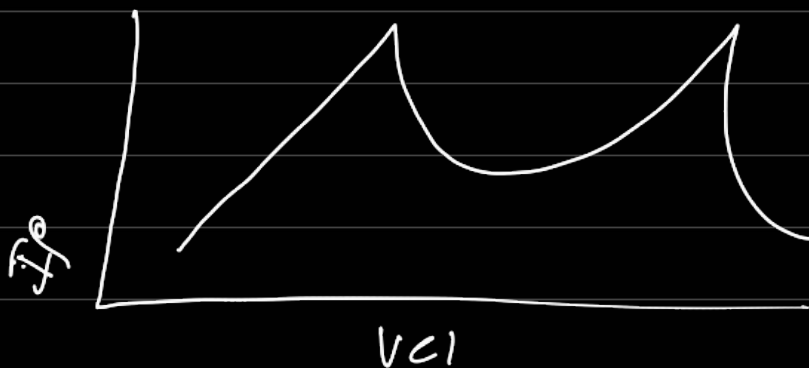
$$c_u \equiv \frac{c_3^2}{G^2}$$

$$L(V) = c_u \left(\frac{V^4}{I_0 \mu^2} \right)$$

(Tully-Fisher)

$$L \propto V_{max}^4$$

• $V = V_{max}$: • $H \approx 21 \text{ cm}$



$$\Delta W = 2 V_{max} \sin(i) \leftarrow \text{Same inclination}$$

Ellipticals:

$$L_{el} = (7.22) I_0 R_e^2$$

Assumption: $L(R) = c_1 I_e R$

Viral:

$$M = c_2 \frac{R \sigma_r^2}{G} \leftarrow \sigma_r^2 = V_{rms}^2$$

• Agnew wanted to find a relationship between

$$R, L, \sigma_r$$

$$\text{Vir: } \frac{2 \left(\frac{1}{2} M_{\text{gal}} \sigma_r^2 \right)}{2K} = \frac{C_2 \left(\frac{G M_{\text{gal}}^2}{R} \right)}{U}$$

$$\sigma_r^2 = C_2 \frac{G M_{\text{gal}}}{R} \quad \tilde{C}_2 = C_2 G$$

$$R = \frac{\tilde{C}_2 M_{\text{gal}}}{\sigma_r^2}$$

• As we did: divide by L : $\frac{M_{\text{gal}}}{L}$

$$\frac{R}{L} = \frac{\tilde{C}_2 \mu}{\sigma_r^2} \Rightarrow L = \frac{\sigma_r^2 R}{\tilde{C}_2 \mu} \quad *$$

$$L = C_2 I_e R^2 \quad *$$

• μ : const.

$$C_1 I_e R^2 = \sigma_r^2 R C_3 \quad C_3 = \tilde{C}_2 \mu$$

$$C_1 I_e R = \sigma_r^2 C_3 \quad C_4 \equiv C_3 / C_1$$

$$I_e R = C_4 \sigma_r^2$$

Take ln:

$$\ln(I_e) + \ln(R_e) = \ln(C_4) + 2 \ln(\sigma_r)$$

Solve for R_e :

$$\ln(R_e) = 2 \ln(\sigma_r) - \ln(I_e) + c$$

Faber - Jackson

$\rightarrow \mu$: constant

• R_e : the radius at which the luminosity is half of the total

